CAPITAL STRUCTURE DETERMINANTS:

Testing Trade-off Theory Against Pecking Order Theory Using SEC Filing Data

An Empirical Analysis of United States Public Companies

Prepared by

Rida Mustafa

July 2026

1. Introduction

Capital structure refers to the manner in which a firm finances its overall operations and growth through a combination of debt and equity. The question of what determines the proportion of debt a firm chooses to carry has remained one of the central topics in corporate finance since the foundational work of Modigliani and Miller. Two competing theoretical frameworks have historically dominated this discussion: the Trade-off Theory and the Pecking Order Theory.

The Trade-off Theory proposes that firms select an optimal level of leverage by weighing the tax advantages associated with debt financing against the potential costs of financial distress. Under this framework, more profitable firms are expected to carry higher levels of debt, since they are better positioned to take advantage of interest tax shields while facing a lower risk of default.

The Pecking Order Theory, in contrast, argues that firms follow a preferential ordering of financing sources. Firms are expected to rely first on internally generated funds, followed by debt, and finally by external equity issuance only as a last resort. This theory predicts the opposite relationship between profitability and leverage: more profitable firms are expected to accumulate greater retained earnings and therefore rely less on external debt financing.

This report presents an empirical investigation into which of these two theories better describes the observed financing behavior of publicly listed United States companies. The analysis uses firm-level financial statement data extracted from filings submitted to the United States Securities and Exchange Commission, covering annual reports filed from 2009 onward.

2. Objectives of the Study

- To construct a firm-year panel dataset of financial variables using raw XBRL data derived from SEC filings.
- To calculate the core capital structure variables required for empirical analysis, namely Leverage, Profitability, Tangibility, Size, and Revenue Growth.
- To examine the direction and strength of the relationship between profitability and leverage through descriptive statistics, correlation analysis, and regression modeling.
- To determine whether the empirical evidence is more consistent with Trade-off Theory or Pecking Order Theory.
- To assess whether leverage patterns differ meaningfully across industry sectors.

3. Dataset Description

The dataset used in this study is titled Financial Statement Data Sets, sourced from Kaggle and originally derived from SEC EDGAR company facts filings. It contains firm-level financial statement line items extracted from United States Generally Accepted Accounting Principles, reported in eXtensible Business Reporting Language format, covering listed United States companies from January 2009 onward.

The raw dataset was provided in three files. The facts.parquet file contained the core financial statement values reported under various standardized tags, and was by far the largest of the three files, comprising 79,448,144 individual line items across 9,641 unique XBRL tags. The submissions.csv file contained filing-level metadata, including company identifiers, ticker symbols, industry classification codes, filing type, and fiscal period end dates. A smaller demonstration file, demo_facts.csv, was also provided as a sample of the structure of the full dataset.

For the purposes of this analysis, six financial statement tags were identified as necessary to construct the required variables: Assets, Liabilities, StockholdersEquity, NetIncomeLoss, PropertyPlantAndEquipmentNet, and Revenues.

4. Data Preparation and Cleaning Methodology

4.1 Filtering to Required Financial Tags

Because the full facts dataset was too large to process efficiently as a whole, a predicate filter was applied at the point of reading the parquet file, so that only rows corresponding to the six required tags were loaded into memory. This substantially reduced the computational burden of the analysis while preserving all information relevant to the study.

4.2 Restriction to Annual Filings

The filtered financial data was merged with the filing metadata contained in submissions.csv using the filing identifier as the merge key. The merged dataset was then restricted to Form 10-K filings only, representing annual reports, since these form the conventional basis for capital structure research. Amended annual filings, designated as Form 10-K/A, were excluded at this stage to avoid duplicated reporting periods.

4.3 Deduplication of Financial Values

Because annual filings routinely report both current and prior-year comparative figures, the same financial value for a given company and fiscal period could appear across multiple filings. To construct a clean company-year panel, duplicate values for the same company, tag, and fiscal period end date were removed, with the most recently filed observation retained on the assumption that later filings are more likely to reflect finalized figures.

4.4 Correction for Embedded Quarterly Values

A further correction was required because some annual filings tag supplementary quarterly figures that do not correspond to the company's official annual reporting period. To address this, the dataset was restricted to observations whose reported period end date matched the filing's official fiscal period end date, ensuring that only genuine annual figures were retained.

4.5 Reshaping to a Wide Company-Year Panel

The cleaned long-format data, in which each row represented a single financial value for a single tag, was reshaped into a wide panel in which each row represented one company-year observation and each column represented one of the six financial variables. This structure was required to compute financial ratios that depend on more than one variable being available for the same observation.

4.6 Recovery of Missing Liabilities

Total Liabilities exhibited a substantial number of missing values in the constructed panel. This reflects a known characteristic of XBRL reporting, in which some filers report only two of the three components of the fundamental accounting identity, namely that Total Assets equal Total Liabilities plus Total Stockholders Equity, leaving the third to be inferred. Missing values for Liabilities were therefore recovered by subtracting Stockholders Equity from Assets wherever both of these values were directly available, without overwriting any values that were already reported directly.

5. Construction of Capital Structure Variables

Following standard definitions established in empirical capital structure research, five variables were constructed from the cleaned panel.

Variable	Definition	Role in Analysis
Leverage	Total Liabilities divided by Total Assets	Dependent variable
Profitability	Net Income divided by Total Assets	Primary explanatory variable
Tangibility	Net Property, Plant and Equipment divided by Total Assets	Control variable
Size	Natural logarithm of Total Assets	Control variable
Revenue Growth	Percentage change in Revenues relative to the prior fiscal year	Control variable

Observations with non-positive Total Assets were excluded prior to computing these ratios, since Total Assets serves as the denominator for most of the constructed variables and a non-positive value would produce an undefined or economically meaningless result.

6. Handling of Implausible Values and Outliers

Following variable construction, several implausible values inconsistent with the economic definitions of the variables were identified and addressed. Infinite values in Revenue Growth, arising from companies with a prior-year revenue of zero, were converted to missing values. Leverage was restricted to a plausible economic range between zero and three, and observations falling outside this range were removed. Profitability, Tangibility, and Revenue Growth were winsorized at the first and ninety-ninth percentiles to limit the influence of extreme outliers. Because Revenue Growth remained highly skewed even after this adjustment, a more aggressive winsorization at the fifth and ninety-fifth percentiles was subsequently applied to this variable alone.

After all cleaning steps, the final analytical panel consisted of 9,733 firm-year observations across approximately 6,600 unique companies.

7. Descriptive Statistics and Distributional Analysis

The distribution of the four core variables, Leverage, Profitability, Tangibility, and Size, was examined using histograms prior to conducting correlation and regression analysis.

- Leverage was concentrated between approximately 0.3 and 0.9, with a pronounced cluster of firms between 0.55 and 0.60, indicating that most firms in the sample finance a moderate to high proportion of their assets through debt.
- Profitability was heavily concentrated near zero, with a long left tail, indicating that while most firms reported small positive or near-zero returns on assets, a subset of firms reported substantial losses relative to their asset base.
- Tangibility was concentrated near zero and decayed sharply, suggesting that a large share of the sampled firms hold relatively few tangible fixed assets, consistent with a sample containing a substantial number of technology, service, and early-stage companies.
- Size followed an approximately bell-shaped distribution centered between 17 and 22 on the natural logarithm scale, indicating that the sample spans a wide range of firm sizes.

8. Correlation Analysis

Pairwise correlations were computed among Leverage, Profitability, Tangibility, Size, and Revenue Growth in order to obtain an initial, purely descriptive indication of the relationship between profitability and leverage before controlling for other factors.

Variable Pair	Correlation Coefficient	Interpretation
Leverage and Profitability	-0.34	Negative relationship, consistent with Pecking Order Theory
Leverage and Size	-0.11	Weak negative relationship
Profitability and Size	0.53	Moderate positive relationship

Variable Pair	Correlation Coefficient	Interpretation
Leverage and Tangibility	0.04	Very weak relationship
Leverage and Revenue Growth	0.03	Very weak relationship

The negative correlation between Leverage and Profitability provides preliminary support for Pecking Order Theory, since it suggests that firms with higher profitability tend to carry lower leverage. The moderate positive correlation between Profitability and Size indicates that these two variables may capture partially overlapping information, a consideration that is addressed jointly in the regression analysis that follows.

9. Regression Analysis

To formally test the relationship between profitability and leverage while controlling for other determinants, two ordinary least squares regression models were estimated, with Leverage as the dependent variable. Model 1 included Profitability, Tangibility, and Size, and was estimated on the largest available sample. Model 2 additionally included Revenue Growth, and was estimated on a smaller subsample for which this variable was available.

Variable	Model 1 Coefficient	Model 2 Coefficient
Profitability	-0.2305 (significant at 1 percent level)	-0.3425 (significant at 1 percent level)
Tangibility	0.1171 (significant at 1 percent level)	Not statistically significant
Size	0.0133 (significant at 1 percent level)	Not statistically significant
Revenue Growth	Not included	Not statistically significant
Sample size	6,679 observations	685 observations
R-squared	0.125	0.244

The coefficient on Profitability is negative and statistically significant at the 1 percent level in both model specifications. This indicates that, holding other variables constant, firms with higher profitability tend to carry lower leverage, a finding that is more consistent with Pecking Order Theory than with Trade-off Theory. The persistence of this negative relationship after including Revenue Growth in Model 2 suggests that the finding is reasonably robust.

Tangibility showed a positive and statistically significant relationship with Leverage in Model 1, consistent with the expectation that tangible assets serve as collateral and increase a firm's capacity to borrow. This effect was not statistically significant in Model 2, which is likely attributable to the considerably smaller sample size available once Revenue Growth is included rather than a genuine change in the underlying relationship.

Size exhibited a small positive and statistically significant coefficient in Model 1, suggesting that larger firms in the sample tend to carry marginally higher leverage, a pattern more consistent with Trade-off

Theory. This effect was smaller and not statistically significant in Model 2. Revenue Growth did not show a statistically significant relationship with Leverage in either specification.

The explanatory power of both models, as measured by R-squared, was modest, at 0.125 for Model 1 and 0.244 for Model 2. This indicates that while Profitability, Tangibility, and Size are statistically meaningful predictors of Leverage, a substantial portion of the variation in leverage across firms remains unexplained by these variables alone, which is consistent with findings in prior empirical capital structure research.

A scatter plot of Profitability against Leverage, with a fitted regression line, visually confirmed the negative relationship identified in the regression analysis. Firms with negative profitability were more widely dispersed across the range of leverage values, while firms with positive profitability tended to cluster at comparatively lower leverage levels.

10. Sector-Level Analysis

To examine whether the relationship between profitability and leverage holds uniformly across industries, firms were grouped into broad sectors based on the first two digits of their Standard Industrial Classification code, and average leverage was compared across these groupings.

- Retail Trade, Agriculture, and Wholesale Trade exhibited the highest average leverage among all sectors, each above 0.72, indicating that firms in these industries finance a large proportion of their assets through debt.
- Manufacturing and Finance, Insurance, and Real Estate exhibited comparatively lower average leverage, below 0.59, despite Manufacturing representing the largest sector in the sample with 3,715 firm-year observations.
- Sectors with the highest average leverage also tended to exhibit the most negative average profitability, reinforcing the overall negative relationship between profitability and leverage, although this pattern was not perfectly uniform across all sectors.
- Finance, Insurance, and Real Estate together with Manufacturing accounted for more than 6,300 of the total firm-year observations, indicating that the overall regression results are substantially shaped by the financing behavior of these two sectors.
- The variation in average leverage across sectors, ranging from approximately 0.54 to 0.76, suggests that industry classification is an important factor in capital structure decisions, beyond the firm-level variables already included in the regression models.

11. Conclusion

This study examined the determinants of corporate capital structure using firm-level financial data derived from SEC filings of United States public companies. The analysis produced a clean panel of 9,733 firm-year observations from a raw dataset of over 79 million individual financial statement line items, through a structured process of filtering, deduplication, correction, and variable construction.

The central finding of this study is a consistently negative and statistically significant relationship between profitability and leverage, observed across both descriptive correlation analysis and two regression model specifications. This finding is more consistent with the predictions of Pecking Order Theory than with those of Trade-off Theory, suggesting that, within this sample, more profitable firms tend to rely more heavily on internal financing and less on external debt.

Tangibility and Size showed evidence of a positive relationship with leverage in the larger sample, consistent with conventional expectations, although these effects lost statistical significance in the smaller sample used to incorporate Revenue Growth. Sector-level analysis further demonstrated that industry classification is an important source of variation in leverage that is not fully captured by firm-level financial characteristics alone.

Overall, the results of this study suggest that, for the sample of United States public companies examined, financing behavior is better explained by the Pecking Order Theory than by the Trade-off Theory, although both firm-specific characteristics and industry membership appear to play a meaningful role in shaping capital structure decisions.

12. Limitations and Future Research

- The dataset used in this study captures only the six financial variables required for the core analysis and does not include information on additional determinants of capital structure discussed in the literature, such as firm age, ownership structure, or market-based measures of leverage.
- The regression models are estimated using ordinary least squares on pooled cross-sectional data rather than a panel data method that accounts for unobserved firm-specific effects, which may influence the estimated coefficients.
- The reduction in sample size when Revenue Growth is included limits the ability to draw firm conclusions about the role of growth in capital structure decisions, and future research could address this limitation through alternative growth measures with broader data availability.
- Future extensions of this work could incorporate fixed effects for industry and time period, alternative leverage measures based on market values, and a longer or more balanced panel to further test the robustness of the findings presented in this report.